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इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते
Chips to Startup Programme
INSTITUTION'S INNOVATION COUNCIL (Ministry of Education Initiative)
BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA
POWERING THE YOUTH EMPOWERING THE NATION

TEAM NAME: FUSION TECH

PROBELM STATEMENT NO: 16

PROBLEM STATEMENT TITLE: Adaptive Reconfigurable Unipolar Binary Neural Network Accelerator for Always-On Edge Keyword Spotting

PS Category: Software

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Real-World Problem

- Edge AI accelerators operate **continuously** in real-world environments.
- SRAM-based weight memory is vulnerable to **Single-Event Upsets (SEUs).**
- Inactive Processing Elements (PEs) still consume **dynamic power** when clocked.

Why It Matters

- A **single-bit memory error** can corrupt BNN inference.
- Unnecessary switching increases **dynamic power consumption.**
- Edge ASICs require **reliability and energy efficiency** simultaneously.

Current Challenges

- Baseline ARe-UBNN-KWS assumes **fault-free weight memory**.
- No **on-the-fly correction** of corrupted weights.
- All **16 PEs** remain active even when activation vectors are zero.
- Fault tolerance must be added **without increasing pipeline stages**.

ARe-UBNN-KWS: Hardened & Power-Optimized Accelerator

Our Solution

Integrate SECDED-based weight correction and fine-grained 16-PE clock gating into the existing ARe-UBNN-KWS datapath without adding pipeline stages.

How It Solves the Problem

- **SECDED:** Detects and corrects single-bit weight errors before computation
- **Fault Detection:** Detects double-bit errors and prevents silent data corruption.
- **Dynamic Clock Gating:** Disables individual PEs when their activation vector is all zero
- **Unmodified Accumulator:** Preserves the baseline inference architecture.
- **Combinational Integration:** SECDED and gating operate within existing pipeline boundaries.
- **ASIC Ready:** Targeted for **SKY130 synthesis and timing validation**.

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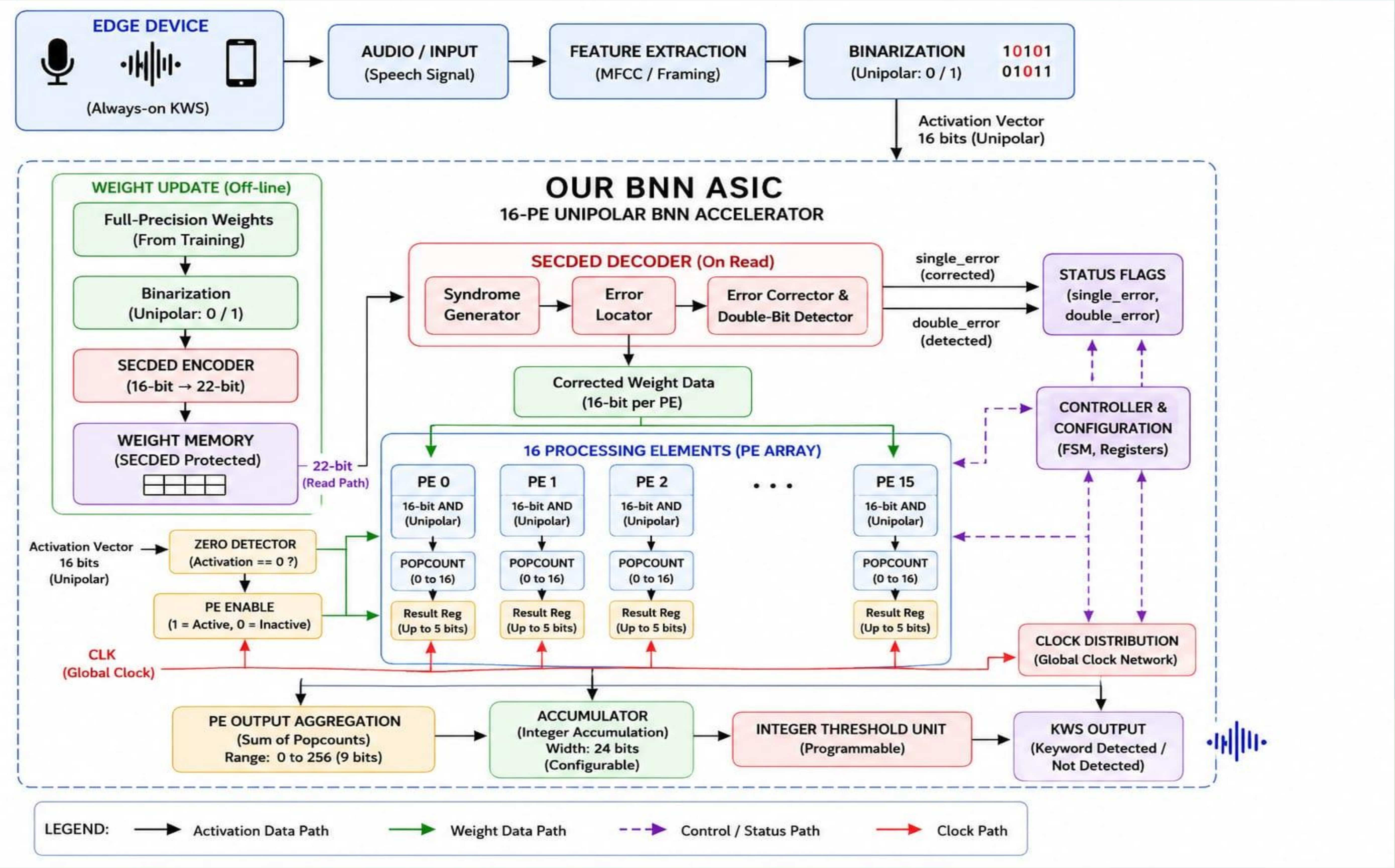

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PURPOSE OF BLOCKS

Input Blocks

- **Audio → Feature Extraction → Binarization**
- Converts speech into **16-bit unipolar activation data**.
- **Trade-off:** Lower computation, but some information is lost during binarization.

Processing Blocks

- **SECDED → Weight Memory → Zero Detector → 16 PEs → Accumulator**
- Provides **fault tolerance** and reduces unnecessary PE switching.
- **Trade-off:** Adds small **area and control overhead**, but improves reliability and power efficiency.

Output Blocks

- **Threshold Unit → KWS Output**
- Converts the accumulated value into the final **keyword decision**.
- **Trade-off:** Small control overhead, but provides **programmable flexibility**.

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Technology & Platform

- Process: SKY130A (130 nm)
- Std Cell: sky130_fd_sc_hd
- Target Clock: 10.0 MHz (100 ns)
- Die Area: 520 x 520 µm
- Core Area: 253,240 um²
- Flow Status: PASS (78/78)
- Top Module:enhanced_wrapper

RTL-to-GDS Stages

- Synthesis: PASS (3,034 cells)
- Floorplan: PASS (PDN Mesh)
- Placement: PASS (7,542 insts)
- CTS: PASS (0.24 ns skew)
- Routing: PASS (162.8 mm wire)
- Post-STA: PASS (+51.1 ns SS)
- Timing Vio: 0 Setup / 0 Hold

Signoff & QualityGDSII

- GENERATED (11.08 MB)
- Magic DRC: 0 errors (CLEAN)
- KLayout: 0 errors (CLEAN)
- Netgen LVS: MATCH (4,204 nets)
- Total Pwr: 1.5732 mW
- IR Drop: 0.2996 mV
- SECDED: 22-bit Corrected

Key Measured SignOff Metrics(Source:output/pnr/metrics.json)

- Die Dimensions: 520 × 520 µm (0.2704 mm² Area)
- Core Utilization: 16.95% (42,929 µm² Active)
- Placed Instances: 7,542 (442 DFFs, 2.5k Comb)
- Setup Worst Slack: +51.10 ns (SS 100°C 1.6V (Slow))
- Hold Worst Slack: +0.062 ns (FF -40°C 1.95V (Fast))
- Estimated Power: 1.5732 mW (10 MHz @ 1.8V, 25°C)
- Worst IR-Drop: 0.2996 mV (<0.02% of 1.8V Rail)
- DRC & LVS: 0 / MATCH (Unique Net Match)

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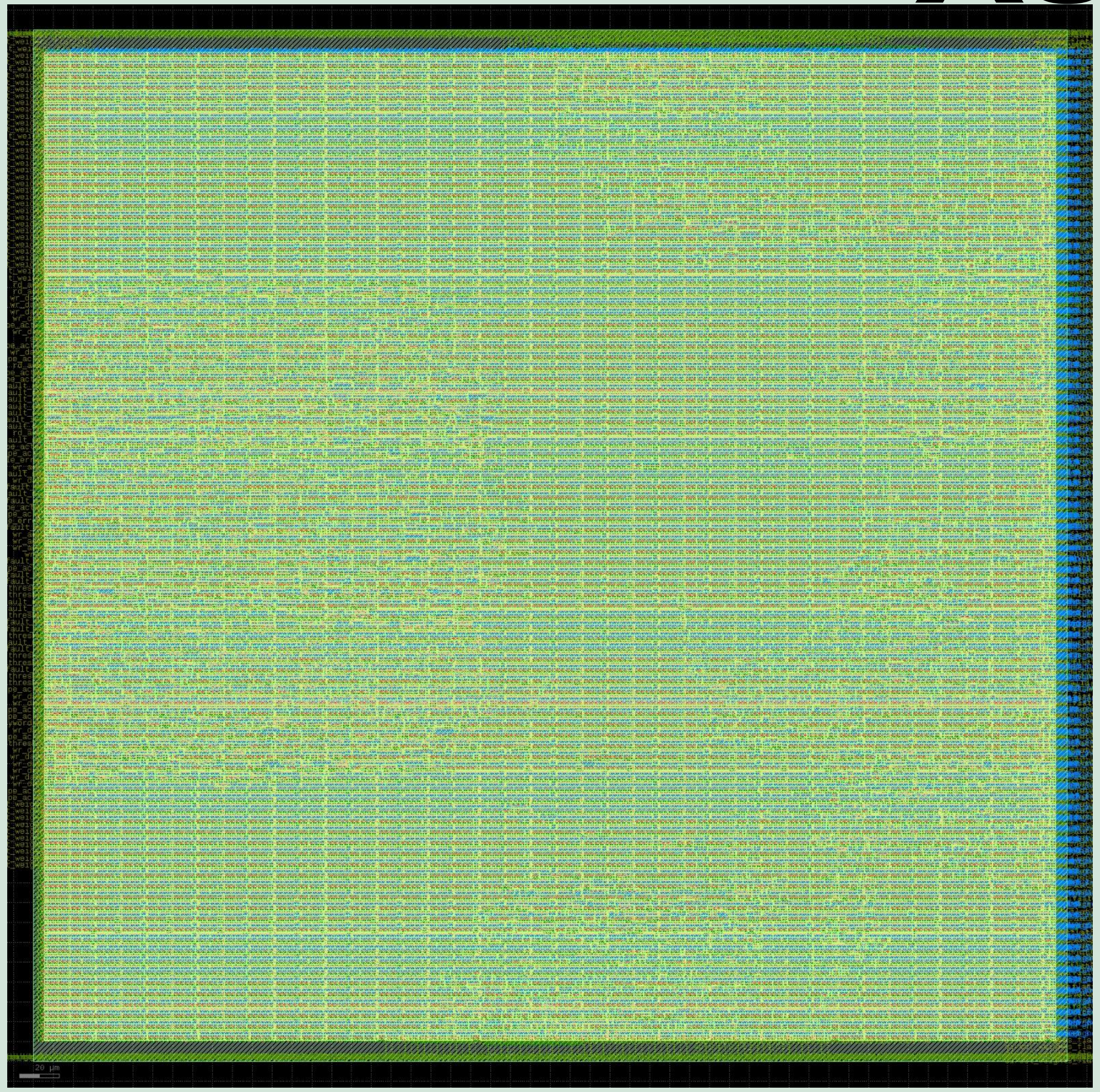
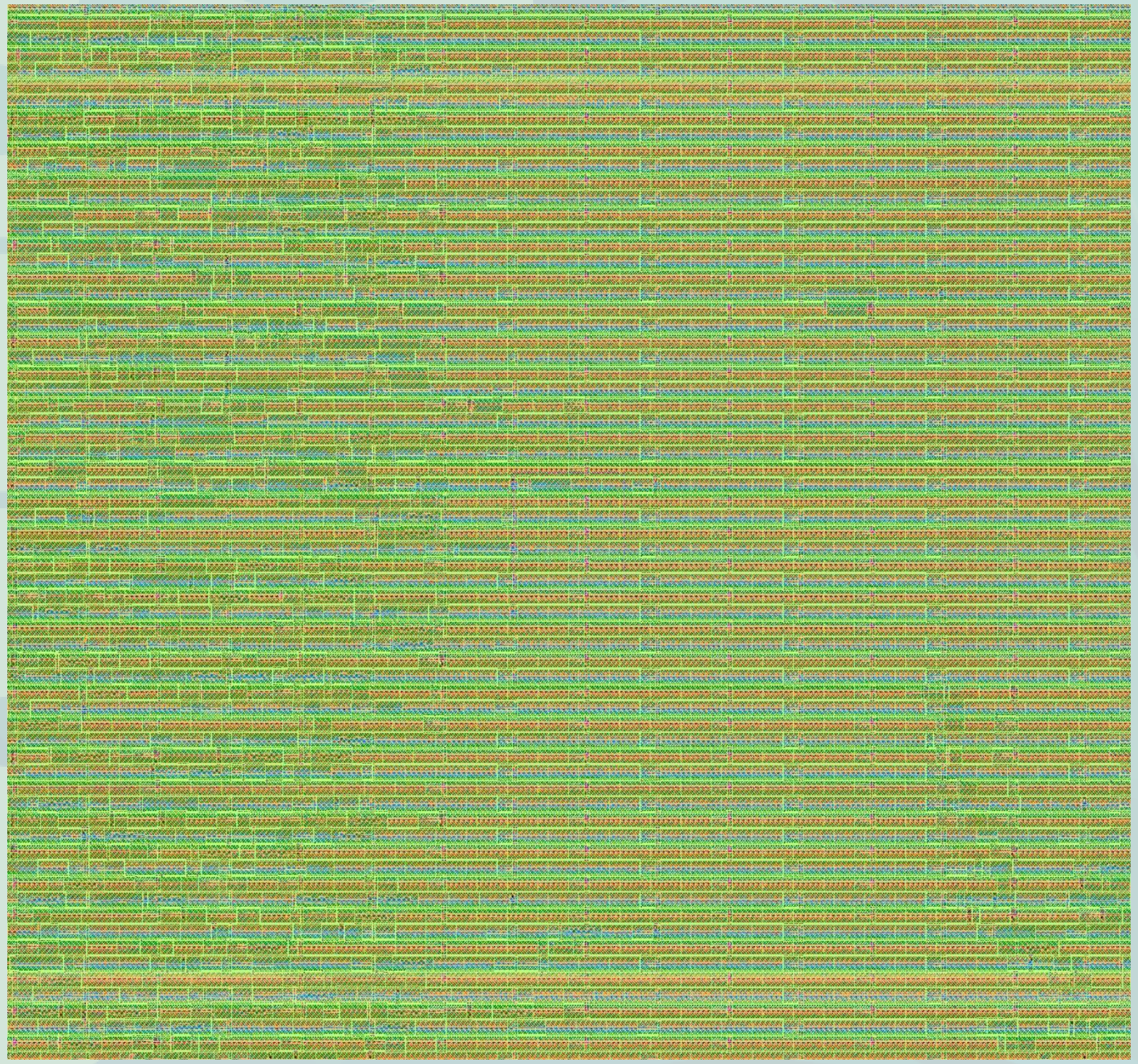
PHYSICAL LAYOUT VIEW OF ASIC DESIGN



FULL DIE



ZOOM - FULL DIE

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```
final_regression_post_pnr.log
File   Edit   View
[PASS] T4_single_bit_fault_bit_4  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_5  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_6  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_7  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_8  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_9  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_10  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_11  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_12  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_13  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_14  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_15  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_16  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_17  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_18  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_19  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_20  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T4_single_bit_fault_bit_21  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T5_double_bit_fault  sum=128 kw=1 SEC=0 DED=1 pe_act=0xffff
[PASS] T6_mixed_sparse  sum=128 kw=1 SEC=0 DED=0 pe_act=0x00ff
[PASS] T7_fault_during_inference  sum=128 kw=1 SEC=1 DED=0 pe_act=0xffff
[PASS] T8_sparse_plus_fault  sum=64 kw=1 SEC=1 DED=0 pe_act=0x5555
[PASS] T9a_all_ones  sum=256 kw=1 SEC=0 DED=0 pe_act=0xffff
[PASS] T9b_all_zeros  sum=0 kw=0 SEC=0 DED=0 pe_act=0x0000
[PASS] T9c_zero_threshold  sum=0 kw=1 SEC=0 DED=0 pe_act=0x0000
[PASS] T10_random_0  sum=47 kw=0 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_1  sum=78 kw=1 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_2  sum=56 kw=0 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_3  sum=73 kw=0 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_4  sum=76 kw=1 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_5  sum=75 kw=1 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_6  sum=52 kw=0 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_7  sum=102 kw=1 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_8  sum=57 kw=1 SEC=0 DED=0 pe_act=0xffff
[PASS] T10_random_9  sum=45 kw=0 SEC=0 DED=0 pe_act=0xffff

================================================
  ARe-UBNN-KWS VERIFICATION SUMMARY
  TOTAL  : 42
  PASSED : 42
  FAILED : 0
================================================
  *** ALL TESTS PASSED ***
================================================
[TB] Done at 3826000 ns
```

RESULT

- ➤ **RTL FUNCTIONAL VERIFICATION RESULTS**
- ➤ **42/42 TESTS PASSED 100% FUNCTIONAL VERIFICATION**
- ➤ Single-bit weight faults corrected on the fly
- ➤ Sparse activations trigger per-PE clock gate
- ➤ Fault-tolerant,power-optimized BNN inference verified succesfully under fault injection and Sparse activation Scenarios

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RESEARCH & REFERENCES References

- IEEE literature on SEU mitigation in SRAM-based memories
- Research on SECDED Hamming Error Correction Codes
- Research on Clock Gating and Low-Power ASIC Design
- Research on Binary Neural Networks and POPCOUNT-based accelerators
- SkyWater SKY130 PDK documentation
- SystemVerilog RTL and ASIC synthesis documentation

Final Statement

- ARe-UBNN-KWS is transformed into a fault-tolerant and sparsity-aware edge-AI datapath while preserving the baseline pipeline architecture.